

AIM:

To visualize the dataset using different plots to identify patterns, distributions, relationships, and anomalies.

PROCEDURE:

1. Import Pandas, NumPy, and Matplotlib libraries.
2. Load the dataset using `pd.read_csv()`.
3. Display the first few records using `head()`.
4. Generate a **Line Chart** to show trends in data.
5. Generate a **Bar Chart** to compare different categories.
6. Generate a **Histogram** to show the distribution of numerical data.
7. Add suitable labels, titles, and legends to the plots.
8. Display the visualizations using `plt.show()`.

PROGRAM:

```
import matplotlib.pyplot as plt

import pandas as pd

import numpy as np

df = pd.read_csv("/content/data.csv")

print(df.head())

# LINE CHART

x = np.linspace(0, 10, 100)

y = np.sin(x)

plt.plot(x, y, label="Sine Wave")

plt.xlabel("X-axis")

plt.ylabel("Y-axis")
```

```
plt.title("Line Plot")
```

```
plt.legend()
```

```
plt.show()
```

```
# BAR CHART
```

```
categories = ['A', 'B', 'C', 'D']
```

```
values = [3, 7, 5, 4]
```

```
plt.bar(categories, values)
```

```
plt.xlabel("Category")
```

```
plt.ylabel("Values")
```

```
plt.title("Bar Plot")
```

```
plt.show()
```

```
# HISTOGRAM
```

```
data = np.random.randn(1000)
```

```
plt.hist(data, bins=20, edgecolor='black')
```

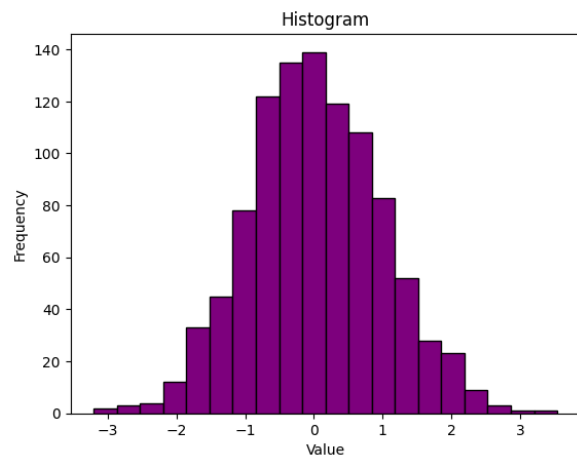
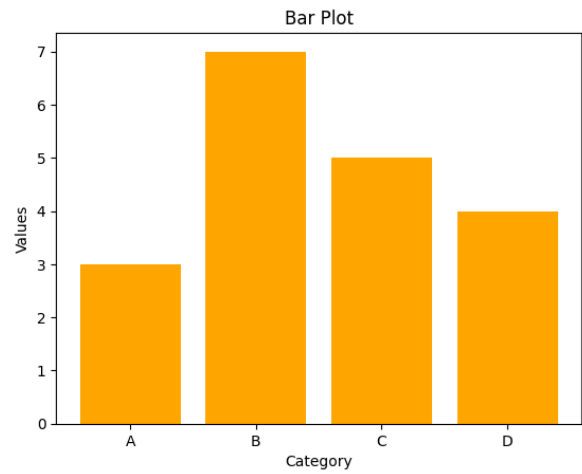
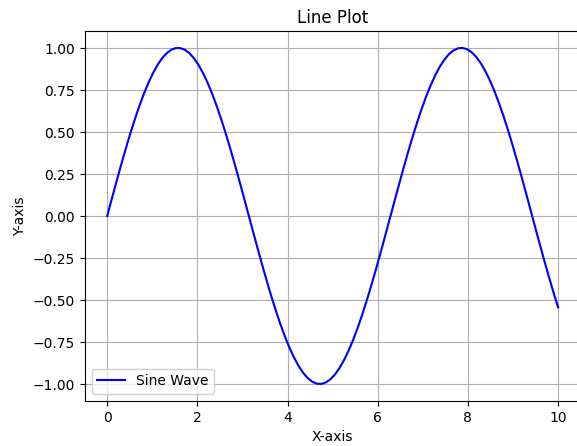
```
plt.xlabel("Value")
```

```
plt.ylabel("Frequency")
```

```
plt.title("Histogram")
```

```
plt.show()
```

OUTPUT:



RESULT:

Thus, the given program was successfully executed and different visualizations were generated to understand patterns and distributions in the data.